

Solution Architecture

EDW-Core Enterprise Data Warehouse

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Author	R. Whitfield — Lead Data Architect
Application owner	Pinku Barman — Group Finance Technology
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Review cycle	Annual, or on material architecture change

Purpose of this document. This document describes the current-state architecture of EDW-Core, the Group Finance enterprise data warehouse. It has been prepared as the baseline input to the GCP cloud onboarding assessment and supersedes version 3.1.

Known limitations. Section 13 records areas where documentation is incomplete. These are being addressed but are not resolved at the time of writing.

VERSION	DATE	AUTHOR	SUMMARY OF CHANGE
3.2	14 Jan 2026	R. Whitfield	Added System D (ServiceNow) integration. Updated volumetrics. Prepared as cloud assessment baseline.
3.1	22 Aug 2025	R. Whitfield	dbt Core adopted for the core transformation layer, replacing 14 stored procedures.
3.0	09 Feb 2025	A. Lindgren	Migration to SQL Server 2019. Consolidation of three regional marts into a single instance.
2.4	17 Mar 2024	A. Lindgren	Tableau Server upgraded to 2023.1. Extract strategy revised.

Solution Architecture

EDW-Core Enterprise Data Warehouse

1. Purpose and scope

EDW-Core is the consolidated reporting warehouse for Group Finance. It ingests data from four upstream systems, applies a layered transformation model, and serves management reporting, regulatory extracts and downstream operational systems.

The platform has been in production since 2019 and currently supports approximately 340 named reporting users across Finance, Treasury and Group Reporting.

1.1 In scope

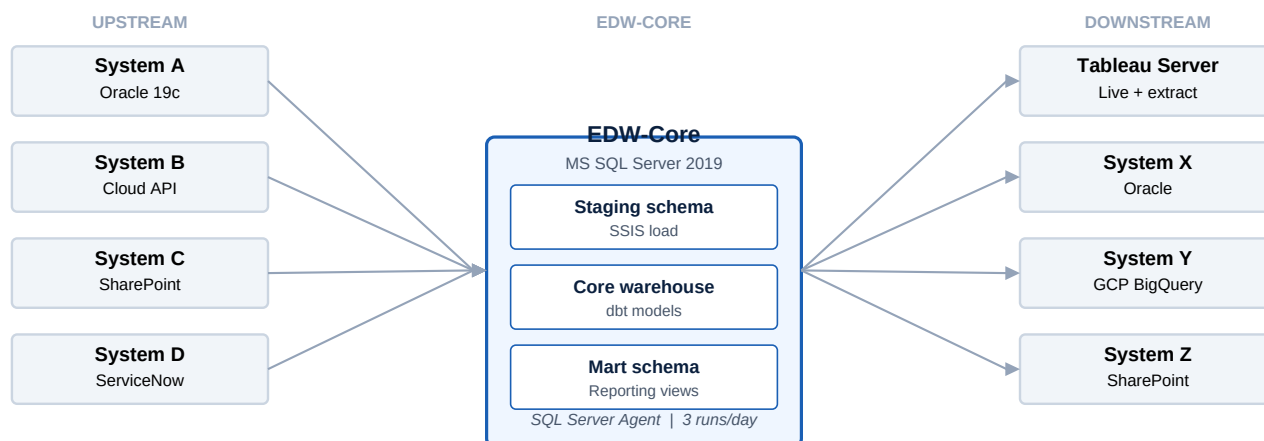
- The EDW-Core database instance and all schemas within it
- All SSIS packages, Python jobs and dbt models that load or transform EDW-Core data
- The SQL Server Agent job schedule and its dependencies
- Four upstream source integrations and four downstream consumers
- The Tableau Server reporting layer where it consumes EDW-Core directly

1.2 Out of scope

- Source system internals — documented separately by their owning teams
- The Group consolidation engine, which reads from System X rather than EDW-Core
- Desktop Tableau workbooks published outside the governed project structure

2. System context

The diagram below shows EDW-Core in relation to its upstream sources and downstream consumers. All components are hosted on-premises in the Frankfurt data centre with the exception of System B and System D, which are external SaaS platforms accessed over the internet, and System Y, which is an existing GCP project.



Solution Architecture

EDW-Core Enterprise Data Warehouse

Figure 1 — EDW-Core system context

2.1 Hosting

EDW-Core runs on a three-node physical cluster in the Frankfurt data centre (rack FRA-D14). The database is a single SQL Server 2019 Enterprise instance in an active/passive failover configuration. SSIS and the Python jobs execute on the same cluster. Tableau Server runs on separate infrastructure in the same facility.

Solution Architecture

EDW-Core Enterprise Data Warehouse

3. Technology stack

LAYER	TECHNOLOGY	VERSION	NOTES
Database	Microsoft SQL Server	2019 Enterprise	Single instance, active/passive failover cluster. Compatibility level 150.
ETL — bulk	SQL Server Integration Services (SSIS)	2019	Approximately 60 packages. Deployed to the SSIS catalogue on the same instance.
ETL — API	Python with Pandas	Python 3.9 / Pandas 1.5	Six scripts handling REST ingestion and file parsing. Executed by SQL Server Agent.
Orchestration	SQL Server Agent	2019	Three master jobs, each chaining SSIS packages and Python steps.
Transformation	dbt Core	1.5.2	94 models across staging, intermediate and mart layers. Runs after the SSIS load completes.
Reporting	Tableau Server	2023.1	Live connections for operational dashboards, scheduled extracts for management reporting.
Version control	Git (Bitbucket)	—	dbt models and Python scripts are versioned. SSIS packages are not.

SSIS packages are not under version control

SSIS packages are deployed directly to the catalogue from developer workstations. There is no repository copy and no formal package inventory. This is a known control weakness and is recorded in section 13.

4. Upstream integrations

EDW-Core ingests from four upstream systems. Interface contracts exist for all four and are held in the Confluence integration space.

SYSTEM	PLATFORM	PATTERN	FREQUENCY	MECHANISM	VOL / RUN
System A	Oracle 19c	Batch extract	Nightly, 01:30	SSIS — OLE DB source	~4.2 GB
System B	Cloud API (REST)	Polling	Hourly	Python / requests	~90 MB
System C	SharePoint Online	File drop	Daily, 06:00	SSIS + Python parser	~35 MB
System D	ServiceNow API (REST)	Polling	Every 4 hours	Python / Pandas	~12 MB

4.1 System A — Oracle

The largest source by volume. A nightly SSIS package extracts 22 tables from the Oracle general ledger, using a watermark column for incremental capture. Full reload is supported but has not been required since March 2024.

Solution Architecture

EDW-Core Enterprise Data Warehouse

The extract window is 01:30 to 02:00 and must complete before the 02:00 master job starts.

4.2 System B — Cloud API

A REST API polled hourly by a Python script. Returns JSON, flattened with Pandas and written to a staging table. Authentication is by API key held in a SQL Server credential. The API is rate limited to 1,000 calls per hour; current usage is well within that.

4.3 System C — SharePoint Online

A daily Excel file published to a SharePoint document library by the Treasury operations team. An SSIS package retrieves it via a mapped drive and a Python step parses the workbook. This integration is fragile — format changes upstream have caused four load failures in the last twelve months.

4.4 System D — ServiceNow

Added in version 3.2. Polls the ServiceNow incident and change tables every four hours to support operational reporting on finance system stability. Lowest volume of the four sources.

Solution Architecture

EDW-Core Enterprise Data Warehouse

5. Downstream consumers

CONSUMER	PLATFORM	INTERFACE	FREQUENCY	MECHANISM
Tableau Server	Tableau 2023.1	Direct SQL and extracts	Live and scheduled	Native SQL Server driver
System X	Oracle	Table push	Nightly, 04:30	SSIS — OLE DB destination
System Y	GCP (BigQuery)	File export	Daily, 05:00	Python — CSV to GCS, then load job
System Z	SharePoint Online	Report file drop	Weekly, Monday 07:00	SSIS — flat file destination

Note that System Y is an existing GCP project operated by the Group Reporting team. EDW-Core already exports a daily curated dataset to BigQuery for their use, so a GCP egress path and service account are already established.

6. Data model and volumetrics

The warehouse follows a three-layer model. Staging holds raw landed data with no transformation. The core layer applies conformed dimensions and business logic through dbt models. The mart layer exposes reporting views consumed by Tableau and downstream systems.

MEASURE	VALUE	NOTES
Total database size	1.0 TB compressed	Page compression enabled on all fact tables
Uncompressed equivalent	~3.8 TB	Estimated from compression ratio sampling
Largest single table	312 GB	FACT_GL_POSTING — 1.4 billion rows
Table count	487	Staging 210, core 194, mart 83
Annual growth	~18 percent	Driven mainly by GL posting volume
Retention	Not formally defined	Current practice retains all history since 2019

Data retention policy is not documented

No formal retention or archival policy exists for EDW-Core. All history since initial load in 2019 is retained online. A policy should be agreed before any migration, since it materially affects target storage sizing and cost.

7. Batch schedule and orchestration

SQL Server Agent runs three master jobs per day. Each job chains SSIS package executions, Python script steps and a final dbt run. Typical end-to-end duration is one hour.

Solution Architecture

EDW-Core Enterprise Data Warehouse

JOB	START	DURATION	SCOPE	CRITICALITY
EDW_MASTER_NIGHTLY	02:00 CET	~62 min	Full load from Systems A and C, all dbt models, System X and Z pushes	Tier 2
EDW_MASTER_MIDDAY	10:00 CET	~48 min	Incremental from Systems B and D, mart layer refresh only	Tier 3
EDW_MASTER_EVENING	18:00 CET	~55 min	Incremental from Systems B and D, full dbt run, System Y export	Tier 2

Job dependencies are not formally mapped

Step ordering within each master job is defined implicitly by the Agent job step sequence. There is no dependency graph or documented DAG structure, so the true precedence relationships between packages are not captured anywhere outside the job definitions themselves.

Solution Architecture

EDW-Core Enterprise Data Warehouse

8. ETL component inventory

8.1 SSIS packages

The SSIS estate is estimated at approximately 60 packages. A documentation exercise was started in October 2025 and has covered 47 of them to date. The remainder are known to exist in the catalogue but have not yet been reviewed or described.

CATEGORY	DOCUMENTED	ESTIMATED TOTAL	NOTES
Source extract — System A	18	18	Complete. One package per Oracle source table group.
Source extract — System C	4	4	Complete.
Staging to core load	16	~22	Six packages identified in the catalogue but not yet documented.
Downstream push	7	7	Complete. Systems X and Z.
Utility and housekeeping	2	~9	Largely undocumented. Includes index maintenance and archive routines.
Total	47	~60	13 packages remain undocumented.

SSIS package inventory is incomplete

47 of approximately 60 packages are documented. The remaining 13 sit mainly in the utility and staging-to-core categories. Completing this inventory is a prerequisite for accurately scoping any migration or rewrite of the ETL layer.

8.2 Python components

SCRIPT	PURPOSE	INVOKED BY	LIBRARIES
ingest_system_b.py	Hourly REST poll and flatten	Agent step, hourly	requests, pandas
ingest_system_d.py	ServiceNow incident and change extract	Agent step, 4-hourly	requests, pandas
parse_treasury_file.py	SharePoint workbook parser	EDW_MASTER_NIGHT LY	openpyxl, pandas
export_system_y.py	Curated export to GCS and BigQuery	EDW_MASTER_EVENING	google-cloud-storage, pandas
reconcile_gl.py	GL control total reconciliation	EDW_MASTER_NIGHT LY	pandas, sqlalchemy
archive_partitions.py	Partition maintenance	Monthly Agent job	pyodbc

Solution Architecture

EDW-Core Enterprise Data Warehouse

9. Transformation layer — dbt

dbt Core 1.5.2 was introduced in version 3.1 of this architecture, replacing 14 stored procedures that previously held the core business logic. Models are versioned in Bitbucket and executed as the final step of each master job.

LAYER	MODELS	MATERIALISATION	NOTES
Staging	38	View	One model per source table. Renaming and type casting only.
Intermediate	31	Ephemeral / table	Conformed dimensions and business rules.
Mart	25	Table	Reporting-ready fact and dimension tables.
Total	94	—	Test coverage: 62 tests, mainly not-null and relationship tests.

dbt lineage has not been exported

dbt generates full model lineage through `dbt docs generate`, but the resulting manifest has never been exported or published. Lineage is therefore only visible to engineers running dbt locally, and cannot be used to validate a migration.

Solution Architecture

EDW-Core Enterprise Data Warehouse

10. Reporting layer

Tableau Server 2023.1 is the sole governed reporting tool for EDW-Core. Two connection patterns are in use.

PATTERN	WORKBOOKS	USERS	NOTES
Live connection	23	~120	Operational dashboards querying the mart layer directly.
Scheduled extract	41	~340	Management reporting. Extracts refresh after each master job.

Extract refresh schedules are not documented centrally

Refresh schedules are configured per workbook within Tableau Server and are not recorded in any central document. Determining the true dependency between the batch schedule and extract refresh currently requires inspecting each workbook individually.

11. Operational characteristics

ASPECT	CURRENT POSITION
Service tier	Tier 2 — Business Important
Availability target	99.5 percent during business hours (07:00–19:00 CET)
RTO	4 hours — agreed with Group Finance in the 2024 service review
RPO	Not formally agreed. Current backup cadence implies approximately 24 hours.
Backup	Nightly full, hourly transaction log. Retained 35 days on disk, 7 years on tape.
Disaster recovery	Passive cluster node in the same facility. No geographically separate DR site.
Monitoring	SQL Server Agent job alerts to a shared mailbox. No proactive data quality monitoring.
Support model	Business hours only. Out-of-hours failures are actioned the following morning.

Failure history over the last twelve months: eleven job failures, of which four were caused by System C file format changes, three by Oracle extract timeouts, two by disk pressure and two by transient network faults. Mean time to restore was 3.2 hours.

12. Security and access

EDW-Core holds internal financial data. No personal data, and no data subject to GDPR or MiFID II reporting obligations, is held in this warehouse. This was confirmed by Data Governance in the March 2025 classification review.

CONTROL	CURRENT POSITION
Data classification	Internal. Confirmed by Data Governance, March 2025.

Solution Architecture

EDW-Core Enterprise Data Warehouse

CONTROL	CURRENT POSITION
Authentication	Windows Authentication via Active Directory.
Authorisation	Three AD groups: EDW_READ, EDW_WRITE, EDW_ADMIN. Applied at database role level.
Encryption at rest	Transparent Data Encryption enabled on the instance.
Encryption in transit	TLS 1.2 enforced for client connections.
Privileged access	Four named DBAs hold sysadmin. Reviewed annually.
Audit logging	SQL Server audit enabled for DDL and login events only.

Access model is coarse-grained and not documented at object level

Permissions are granted through three broad AD groups at database level. There is no documented mapping of which teams should access which schemas or tables, and no row- or column-level controls. Any target-state IAM model will need to be designed from scratch rather than translated.

Solution Architecture

EDW-Core Enterprise Data Warehouse

13. Known documentation gaps

The following gaps are known at the time of writing. They are recorded here so that any downstream assessment is not misled by their absence.

REF	GAP	IMPACT	OWNER	TARGET
G-01	13 of approximately 60 SSIS packages are undocumented	ETL migration cannot be accurately scoped	Data Engineering	Q1 2026
G-02	dbt lineage manifest never exported	No machine-readable dependency graph for validation	Analytics Engineering	Q1 2026
G-03	SQL Server Agent job dependencies not mapped to a DAG	Precedence relationships exist only in job definitions	Data Engineering	Q1 2026
G-04	Tableau extract refresh schedules not centrally documented	Batch-to-refresh dependency unclear	BI Team	Q2 2026
G-05	No formal data retention or archival policy	Storage sizing and cost cannot be modelled accurately	Data Governance	Q2 2026
G-06	Object-level access model undocumented	Target IAM design must start from first principles	Security	Q1 2026
G-07	RPO not formally agreed with the business	Backup and DR design cannot be validated	Service Management	Q2 2026

Appendix A — Infrastructure inventory

HOST	ROLE	SPECIFICATION	LOCATION
FRA-SQL-014	Primary database node	32 vCPU, 256 GB RAM, 2 TB SSD	Frankfurt, rack FRA-D14
FRA-SQL-015	Passive failover node	32 vCPU, 256 GB RAM, 2 TB SSD	Frankfurt, rack FRA-D14
FRA-ETL-009	SSIS and Python execution	16 vCPU, 64 GB RAM, 500 GB SSD	Frankfurt, rack FRA-D15
FRA-TAB-003	Tableau Server	16 vCPU, 128 GB RAM, 1 TB SSD	Frankfurt, rack FRA-E02
SAN-FRA-02	Shared storage	4 TB allocated of 40 TB array	Frankfurt

Appendix B — Glossary

TERM	DEFINITION
EDW	Enterprise Data Warehouse
SSIS	SQL Server Integration Services — Microsoft ETL tooling

Solution Architecture

EDW-Core Enterprise Data Warehouse

TERM	DEFINITION
dbt	Data build tool — SQL-based transformation framework
Mart	Reporting-ready subject area exposed to consumers
Watermark	Column used to identify records changed since the last extract
TDE	Transparent Data Encryption
NAR ID	Application register identifier

End of document. Prepared by R. Whitfield, Lead Data Architect, Group Finance Technology. For queries contact the Data & Analytics architecture team.